From the nominal cosh to the fitted cosh: what the minimiser absorbs, step by step

every intermediate carried; nothing skipped

The question is: if the true signal is the nominal response plus a deviation e_ω, and the minimiser fits a cosh to the sum, what is left over? The answer decides what a fit residual can and cannot tell us, so it is worth doing slowly. Sections 1–3 set up e_ω, Sec. 4 is the one piece of algebra everything turns on, and Secs. 5–8 apply it.

1 The nominal response, from the dynamics

Over the contact phase the linearised tip-over dynamics are

$$J_P\ddot{\varphi} = Wz_{CoM}\,\varphi + \big(M(t) - M_{\text{crit}}\big), \tag{1}$$

the sign on Wz_{CoM} being positive because past the balance point gravity *drives* the tip rather than restoring it. Measure time from the onset, $\tau = t - t_{\text{c}}$, so that $M(t) - M_{\text{crit}} = \dot{M}\tau$, and divide by J_P:

$$\ddot{\varphi} - C_2^2\varphi = \frac{\dot{M}\tau}{J_P}, \qquad C_2^2 \triangleq \frac{Wz_{CoM}}{J_P}. \tag{2}$$

Particular solution. Try $\varphi_p = \alpha\tau$. Then $\ddot{\varphi}_p = 0$ and (2) gives $-C_2^2\alpha\tau = \dot{M}\tau/J_P$, so

$$\alpha = -\frac{\dot{M}}{J_P C_2^2} = -\frac{\dot{M}}{Wz_{CoM}}, \qquad \text{i.e.} \qquad \varphi_p = -\frac{\dot{M}}{Wz_{CoM}}\,\tau.$$

Homogeneous solution. $\ddot{\varphi}_h = C_2^2\varphi_h$ gives $\varphi_h = A\cosh C_2\tau + B\sinh C_2\tau$. So

$$\varphi(\tau) = A\cosh C_2\tau + B\sinh C_2\tau - \frac{\dot{M}}{Wz_{CoM}}\tau. \tag{3}$$

Initial conditions. At the onset the vehicle is at rest at the balance point: $\varphi(0) = 0$ and $\dot{\varphi}(0) = 0$. The first gives $A = 0$. Differentiating, $\dot{\varphi} = BC_2\cosh C_2\tau - \dot{M}/Wz_{CoM}$, and at $\tau = 0$,

$$BC_2 - \frac{\dot{M}}{Wz_{CoM}} = 0 \quad\Longrightarrow\quad B = \frac{\dot{M}}{Wz_{CoM}C_2}.$$

The rate. Substituting A and B and differentiating once,

$$\boxed{\omega_{\text{nom}}(\tau) = \dot{\varphi} = \frac{\dot{M}}{Wz_{CoM}}\big(\cosh C_2\tau - 1\big) = C_1\big(\cosh C_2\tau - 1\big), \qquad C_1 \triangleq \frac{\dot{M}}{Wz_{CoM}} = K\dot{M}.} \tag{4}$$

Two things to notice, because both matter later. The amplitude C_1 is *not free*: it is fixed by the ramp rate and the rig constant K. And $\omega_{\text{nom}}(0) = 0$, so the curve leaves the onset flat.

2 The deviation, when an unmodelled moment acts

Let $\rho(\tau)$ be a moment the model does not contain. The true tilt obeys (1) with ρ added:

$$J_P\ddot{\varphi} = Wz_{CoM}\varphi + \dot{M}\tau + \rho(\tau).$$

Write $\varphi = \varphi_{\text{nom}} + e_\varphi$ and subtract (2). Everything the two have in common cancels, leaving

$$\ddot{e}_\varphi - C_2^2 e_\varphi = \frac{\rho(\tau)}{J_P}, \qquad e_\varphi(0) = \dot{e}_\varphi(0) = 0, \tag{5}$$

with zero initial conditions because the nominal solution already satisfies them.

Impulse response. For $\ddot{y} - C_2^2 y = \delta(\tau)$ with $y(0) = \dot{y}(0) = 0$ the solution is $h(\tau) = \sinh(C_2\tau)/C_2$: it vanishes at $\tau = 0$, its derivative $\cosh C_2\tau$ equals 1 there, which is the unit jump the impulse delivers, and it satisfies the homogeneous equation away from the origin.

Duhamel. Hence

$$e_\varphi(\tau) = \frac{1}{J_P}\int_0^\tau \frac{\sinh C_2(\tau - s)}{C_2}\,\rho(s)\,ds. \tag{6}$$

Differentiate with respect to τ. The Leibniz rule gives a boundary term plus an integral,

$$\dot{e}_\varphi = \underbrace{\frac{\sinh C_2(\tau-\tau)}{C_2 J_P}\rho(\tau)}_{=0} + \frac{1}{J_P}\int_0^\tau \cosh C_2(\tau - s)\,\rho(s)\,ds,$$

and the boundary term vanishes because $\sinh 0 = 0$. So

$$\boxed{e_\omega(\tau) = \frac{1}{J_P}\int_0^\tau \cosh\big(C_2(\tau - s)\big)\rho(s)\,ds.} \tag{7}$$

3 Splitting the kernel

The addition formula $\cosh(u-v) = \cosh u\cosh v - \sinh u \sinh v$ applies to the kernel with $u = C_2\tau$ and $v = C_2 s$:

$$\cosh C_2(\tau - s) = \cosh C_2\tau\,\cosh C_2 s - \sinh C_2\tau\,\sinh C_2 s.$$

The factors in τ do not depend on s, so they come out of the integral:

$$e_\omega(\tau) = \frac{1}{J_P}\Big[\cosh C_2\tau\;P(\tau)\;-\;\sinh C_2\tau\;Q(\tau)\Big], \tag{8}$$

$$P(\tau) \triangleq \int_0^\tau \cosh C_2 s\;\rho(s)\,ds, \qquad Q(\tau) \triangleq \int_0^\tau \sinh C_2 s\;\rho(s)\,ds. \tag{9}$$

This is exact — no approximation yet. P and Q are the running moments of ρ against the two kernels; if ρ has died away by some time, they stop growing and become constants from there on.

4 Two hyperbolics make one

Claim. $a\cosh u + b\sinh u = R\cosh(u + \psi)$ with $R = \sqrt{a^2 - b^2}$ and $\tanh\psi = b/a$, valid when $|a| > |b|$.

Proof. Look for R, ψ with $a = R\cosh\psi$ and $b = R\sinh\psi$. Then

$$R\cosh(u+\psi) = R\big[\cosh u\cosh\psi + \sinh u \sinh\psi\big] = (R\cosh\psi)\cosh u + (R\sinh\psi)\sinh u = a\cosh u + b\sinh u,$$

which is the claim. It remains to check that such R, ψ exist. Squaring and subtracting,

$$a^2 - b^2 = R^2\bigl(\cosh^2\psi - \sinh^2\psi\bigr) = R^2,$$

so $R = \sqrt{a^2 - b^2}$, real exactly when $|a| > |b|$; and dividing, $b/a = \tanh\psi$, which has a unique solution ψ because tanh is a bijection from $\mathbb{R}$ to $(-1, 1)$ and $|b/a| < 1$. □

Read it as a time shift. Since $u = C_2\tau$,

$$R\cosh(C_2\tau + \psi) = R\cosh\Bigl(C_2\bigl(\tau + \psi/C_2\bigr)\Bigr),$$

so adding a sinh to a cosh does not produce a new shape at all: it produces the *same* cosh, with the amplitude changed from a to R and the time origin moved by $-\psi/C_2$.

5 The measured signal is again a cosh

Add the nominal (4) and the deviation (8), and write b_0 for the true pre-onset level:

$$\begin{aligned} y(\tau) - b_0 &= \underbrace{C_1\cosh C_2\tau - C_1}_{\text{nominal}} + \underbrace{\frac{P}{J_P}\cosh C_2\tau - \frac{Q}{J_P}\sinh C_2\tau}_{e_\omega} \\ &= \underbrace{\Bigl(C_1 + \frac{P}{J_P}\Bigr)}_{a}\cosh C_2\tau + \underbrace{\Bigl(-\frac{Q}{J_P}\Bigr)}_{b}\sinh C_2\tau \;-\; C_1. \end{aligned} \tag{10}$$

Wherever P and Q vary slowly enough to be treated as constants, Sec. 4 applies verbatim:

$$\boxed{y(\tau) - b_0 = R\cosh\Bigl(C_2\bigl(\tau - \delta\bigr)\Bigr) - C_1, \qquad R = \sqrt{a^2 - b^2}, \qquad \delta = -\frac{\psi}{C_2}, \qquad \tanh\psi = \frac{b}{a}.} \tag{11}$$

The perturbed signal is exactly a cosh. Not approximately, not to leading order — exactly, with a different amplitude and a different onset.

6 What the minimiser can absorb

The estimator fits

$$\hat{\omega}(\tau) = \hat{C}_1\Bigl(\cosh\bigl(C_2(\tau - \hat{\delta})\bigr) - 1\Bigr) + C, \tag{12}$$

with C the baseline. Expand the shifted cosh by the addition formula:

$$\hat{\omega} = \hat{C}_1\cosh(C_2\hat{\delta})\,\cosh C_2\tau - \hat{C}_1\sinh(C_2\hat{\delta})\,\sinh C_2\tau - \hat{C}_1 + C. \tag{13}$$

Now compare (13) with the data (10), coefficient by coefficient. There are three independent shapes present — $\cosh C_2\tau$, $\sinh C_2\tau$ and the constant 1 — so three equations:

shape	data	fit
$\cosh C_2\tau$	$a = C_1 + P/J_P$	$\hat{C}_1\cosh C_2\hat{\delta}$
$\sinh C_2\tau$	$b = -Q/J_P$	$-\hat{C}_1\sinh C_2\hat{\delta}$
1	$-C_1 + b_0$	$-\hat{C}_1 + C$

Case A: amplitude, onset and baseline all free. Three unknowns $(\hat{C}_1, \hat{\delta}, C)$, three equations, and they are solvable — take $\hat{C}_1 = R$, $\hat{\delta} = \delta$ from (11), and $C = b_0 - C_1 + R$. *The fit is exact and the residual is identically zero.* Not small: zero. A fit with both shape parameters free cannot see e_ω at all, because e_ω has moved the signal along the model family rather than off it.

Case B: the amplitude pinned, $\hat{C}_1 = C_1 = K\dot{M}$. Now there are two unknowns for three equations, and the system is overdetermined. *A least-squares fit does not respond to that by satisfying two rows and calling the third a residual* — it trades across all three to minimise the norm, and it will happily accept error in the sinh row to reduce a larger error in the cosh row. The solution below therefore describes coefficient *matching*, which is a different estimator from the one deployed; §7 gives the difference and it is not small. Matching the sinh row fixes the onset,

$$\sinh C_2\hat{\delta} = -\frac{b}{C_1} = \frac{Q}{J_P C_1}, \tag{14}$$

and the constant row fixes C. The cosh row is then left over, and what it cannot match is the residual:

$$\text{residual coefficient on } \cosh C_2\tau \;=\; a - C_1\cosh C_2\hat{\delta} \;=\; C_1 + \frac{P}{J_P} - C_1\sqrt{1+\Big(\frac{Q}{J_P C_1}\Big)^2}, \tag{15}$$

using $\cosh = \sqrt{1+\sinh^2}$ with (14). For a small deviation the square root is $1 + O(Q^2)$ and this collapses to

$$\boxed{\text{residual} \;\simeq\; \frac{P(\tau_{\text{end}})}{J_P}\cosh C_2\tau \;+\; O(Q^2).} \tag{16}$$

So pinning the amplitude is what makes anything observable, and what survives is the cosh half of the deviation — the P half. The sinh half, the Q half, has gone entirely into the onset.

7 What went into the onset, and what it costs

Solve (14) for small $\hat{\delta}$, where $\sinh C_2\hat{\delta} \simeq C_2\hat{\delta}$:

$$\hat{\delta} \simeq \frac{Q(\tau_{\text{end}})}{J_P C_1 C_2}. \tag{17}$$

The identified threshold is the moment at the identified onset, and the moment ramps at $\dot{M}$, so a shift $\hat{\delta}$ costs $\Delta M_{\text{crit}} = \dot{M}\hat{\delta}$. Substituting $C_1 = \dot{M}/Wz_{CoM}$ and then $Wz_{CoM} = J_P C_2^2$,

$$\Delta M_{\text{crit}} = \dot{M}\cdot\frac{Q}{J_P C_2}\cdot\frac{Wz_{CoM}}{\dot{M}} = \frac{Q\,Wz_{CoM}}{J_P C_2} = \frac{Q\,J_P C_2^2}{J_P C_2} = C_2\,Q(\tau_{\text{end}}) = C_2\int_0^{\tau_{\text{end}}}\sinh(C_2 s)\,\rho(s)\,ds. \tag{18}$$

The ramp rate has cancelled twice over: once because $C_1 \propto \dot{M}$ and once because $\Delta M_{\text{crit}} = \dot{M}\hat{\delta}$. What reaches the threshold is a property of the rig and of ρ, not of how fast the run was driven.

The step from $\sinh C_2\hat{\delta}$ to $C_2\hat{\delta}$ is the only approximation in the section. Keeping the exact root $\hat{\delta} = \operatorname{arcsinh}(z)/C_2$ with $z = Q/(J_P C_1)$ and expanding $\operatorname{arcsinh} z = z - z^3/6 + \ldots$, the term dropped is $\dot{M}z^3/(6C_2)$ — on a representative run 9.2×10^{-6} mN m against the 1.52 of the leading term, so it never matters here, but it is what a numerical check will find if the two forms are compared to machine precision (`analysis/absorption_check.py`).

(18) **is not what the estimator computes.** It was obtained by matching coefficients. The deployed estimator does least squares over the whole window, and picks

$$\hat{\delta} = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2}, \qquad \chi \triangleq \frac{\partial\hat{\omega}}{\partial t_c} = -C_1 C_2\sinh C_2\tau, \tag{19}$$

the projection of the deviation onto the one direction the onset can move it. These do not merely differ in the tail — *they can carry opposite signs*, and the reason is the 3.4° of §8(v). Take ρ an impulse of mass m at s, so that $P = m\cosh C_2 s$ and $Q = m\sinh C_2 s$ exactly and P, Q are constant thereafter: the early-ρ limit in which (18) was supposed to be exact. It gives $\Delta M_{\mathrm{crit}} = +C_2 m \sinh C_2 s$, the onset moving *later*. Measured on the deployed estimator at $s = 77$ ms and $\dot{M} = 1.20$ N m/s, the onset moves 26 ms *earlier*, a threshold error of -31.7 mN m against the $+20.2$ predicted by (18).

The mechanism is plain once stated. A positive ρ raises the amplitude, $R > C_1$; with C_1 pinned the fit cannot follow, and the only way to make a fixed-amplitude cosh larger is to start it earlier. That pull is proportional to P and it overwhelms the $+Q$ term from the sinh row. Keeping it,

$$\Delta M_{\mathrm{crit}} \;\simeq\; C_2\Big[\,Q \;-\; \frac{A(x)}{B(x)}\,P\,\Big], \qquad \begin{array}{l} A = \frac{1}{4}\cosh 2x - \cosh x + \frac{3}{4}, \\ B = \frac{1}{4}\sinh 2x - \frac{1}{2}x, \end{array} \tag{20}$$

with $A/B = 0.85$ at $x = 3.1$ and $\to 1$ for large x. On the impulse above this returns -29.3 mN m against the measured -31.7, an 8% agreement where (18) had the sign wrong. Equation (20) still assumes P and Q have stopped growing; when ρ acts across the whole window neither form applies and only (19) does. **The manuscript uses** (19) **throughout** — it is that form which produces the normalised weight w with $\int w = 1$, and (107), (108) and the reported bound are unaffected by anything in this paragraph. What is affected is the story told about them, below.

8 Three consequences, and the numbers

(i) The residual and the threshold error are complementary. By construction the least-squares onset makes the residual orthogonal to χ. So the residual is everything *orthogonal* to the estimator's span, and ΔM_{crit} is exactly the χ-projection the residual has thrown away. Neither carries any information about the other, and comparing a fit residual against a bound on ΔM_{crit} tests nothing.

(ii) An impulse at the onset moves the onset MOST. This is the reverse of what (18) suggests and of what an earlier draft of this section claimed. If ρ acts only at $\tau \approx 0$ then $Q \approx 0$ because $\sinh 0 = 0$, so the sinh row is silent — but $P \approx \int \rho$ is at its largest, and by (20) the P term carries nearly the same coefficient. The normalised weight of (107) settles it without approximation:

$$w(0) = \frac{C_2 \sinh^2 x}{2\big[\frac{1}{4}\sinh 2x - \frac{1}{2}x\big]} = 5.16\ \mathrm{s}^{-1} \quad \text{against a mean of 1.63,}$$

and w falls monotonically to 0.31 at the window end. Sweeping a narrow bump across the window and fitting with the deployed estimator reproduces $w(s)$ to within 11% over the middle of the range. So the Chebyshev pairing in the manuscript is favourable *because* the physical channels vanish at the onset and rise, not because late forcing is the dangerous kind. Reverse the arrangement — same window mean $\bar{\rho}$, but falling instead of rising — and the threshold moves by 1.4 to 1.6 $\bar{\rho}$, outside the bound. The monotonicity hypothesis is load-bearing.

(iii) Releasing C_1 would destroy the measurement. Case A says the residual would be zero, so the fit would be free to wander along the family. Measured on these runs, releasing amplitude and onset together drives the onset to ± 150 ms — the limit imposed on the search — at every ramp rate, with the amplitude moving from -17 to $+44$%. At $\dot{M} = 1.20$ N m/s, 150 ms of onset is 180 mN m of threshold, fifteen times the entire a priori budget.

(iv) The fitted curve stays in the band the data is in. Since C_1 is pinned, the only freedom left is δ, and $\hat{\omega} - \omega_{\mathrm{nom}} = -2C_1 \sinh \frac{C_2\delta}{2} \sinh C_2(\tau - \frac{\delta}{2})$ exactly. To first order that is

$C_1C_2|\delta|\sinh C_2\tau$, and with $|\delta| \le \bar\rho/\dot M$ and $Wz_{CoM} = J_PC_2^2$ it equals $\bar\rho\sinh(C_2\tau)/(J_PC_2)$ — the same envelope that bounds e_ω itself. So the minimiser returns a cosh lying somewhere between $\omega_{\mathrm{nom}} \pm \bar\rho\sinh(C_2\tau)/(J_PC_2)$, and the band edges are attained at the extreme δ, so the band is the reachable set rather than a container. This is (i) read geometrically: $|\Delta M_{\mathrm{crit}}| \le \bar\rho$ and "the fit lies in the band" are the same inequality. Derived in full, with the second-order term kept, in `docs/vi_e_derivation.tex`, (D.13d)–(D.13g).

(v) Everything above is one angle. Remove the constant — the baseline fits it — and normalise. Over $x = C_2\tau_{\mathrm{end}} = 3.1$ the correlation between $\cosh C_2\tau$ and $\sinh C_2\tau$ is 0.99824: they are 3.4° apart. That single number carries the section. It is why releasing C_1 destroys the measurement (the joint problem has condition number $1/\sin\theta = 16.9$); why pinning C_1 is nonetheless cheap in variance; why an amplitude error masquerades as an onset shift rather than as a residual; why $w(0)$ is the maximum and not zero; and why the residual is a poor witness — a perturbation along cosh loses 99.8% of itself to the estimator's own directions before any of it can reach the residual. The angle widens as the window shortens, so a shorter fit window separates the two channels better at the cost of signal.

(vi) The calibration stage absorbs as well. The onset sweep holds C_1 and C_2 fixed, so shift and baseline are the only channels open to it — but the per-configuration PNLS fit that *supplies* C_2 and K has both free, and it is fitted to runs that already contain ρ. Whatever ρ looks like to a free exponent is taken up by C_2; whatever it looks like to a free amplitude is taken up by K; and the onset sweep then treats the result as truth. Measured on synthetic runs carrying the physical channels, the fitted C_2 comes out 1.3–1.5% high — ρ_φ is destabilising, so the vehicle tips faster than the linear model and a free exponent absorbs the difference — and K comes out 5 to 12% low. Pinning *those* constants and sweeping the onset costs a further 1.1 to 3.0 mN m of threshold, which is the same order as the direct path this document is about. Two remarks. The C_2 excess here is a small part of the 12.3% by which some fitted values exceed their geometric ceiling, so it is a contributor and not the explanation — an overstated CAD J_P remains the likelier bulk. And the C_2 and K biases partly cancel in the onset, which is why the cost is 3.0 rather than the 10 mN m a 5% amplitude error alone would give.

quantity	what it is	measured
$P/(J_PC_1)$	the cosh half, (16)	1.7% median, 3.9% at p90
C_2Q	the sinh half, (18)	0.049 mN m median, 0.93 worst
	against the a priori bound	12.04 mN m
onset if C_1 released	Case A	±150 ms, i.e. 180 mN m
span projection, one run	what the estimator absorbs	0.555 → 0.010 mN m

Reproduced by `analysis/amplitude_channel.py` for the first row, `analysis/error_budget.py` and `analysis/bound_validation_figure.py` for the rest.